of related Linux containers. Pods are scheduled across nodes, which are physical or virtual machines that make up the cluster. A service is implemented by one or more pod replicas.

- A security context can be applied to pod deployments to ensure that the container runs as a non-root user with limited privileges. A pod security policy can be applied to the cluster to enforce that no container is allowed elevated privileges.
- When an API makes network requests to a URL provided by a user, you should ensure that you validate the URL to prevent SSRF attacks. Strict whitelisting of permitted URLs should be preferred to blacklisting. Ensure that redirects are also validated. Protect your APIs from DNS rebinding attacks by strictly validating the Host header and enabling TLS.
- Enabling TLS for all internal service communications protects against a variety of attacks and limits the damage if an attacker breaches your network. A service mesh such as Linkerd or Istio can be used to automatically manage mTLS connections between all services.
- Kubernetes network policies can be used to lock down allowed network communications, making it harder for an attacker to perform lateral movement inside your network. Istio authorization policies can perform the same task based on service identities and may be easier to configure.
- A Kubernetes ingress controller can be used to allow connections from external clients and apply consistent TLS and rate-limiting options. By adding the ingress controller to the service mesh you can ensure